Supporting Information:

Supporting Information: Condition identity and label weighting govern structure–response inference from reused site-resolved PTM data

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1 Protein-level comparison with bulk responses

The source study reports bulk CML-ELISA responses for casein, haemoglobin and collagen at three modification levels each. Figure S1 summarises the comparison discussed in the main text.

The observed ordering is stable: casein exceeds haemoglobin, which exceeds collagen, at all three levels (Figure S1A), so the comparison does not fail on the measurement side. One value, casein at the low modification level, exceeds unity (1.34, SD 0.61) and is therefore not physically interpretable as a repair fraction; we retain it as reported and note that it does not affect the ordering.

Across the 96 of 192 pipeline configurations in which all three scaffolds were represented, seven recovered the observed ordering (Figure S1B), against a chance expectation of one in six. The two most frequent predicted orderings place haemoglobin or collagen first. These

configurations are nested variants of a single pipeline and are strongly correlated, so the fraction is descriptive and not a significance test.

The prespecified configuration could not be evaluated at all because collagen contributed no lysine meeting both the structure-confidence and applicability-domain criteria (Figure S1C). Of 69 collagen lysines, none met the strict structure-quality criterion and one fell inside the applicability domain; for casein the corresponding counts were 8 and 18 of 58, and for haemoglobin 24 and 12 of 24.

With three scaffolds, an exact-ordering test cannot yield a P value below one in six, so this comparison could not have supported a transfer claim under any outcome. It is reported as an applicability stress test.

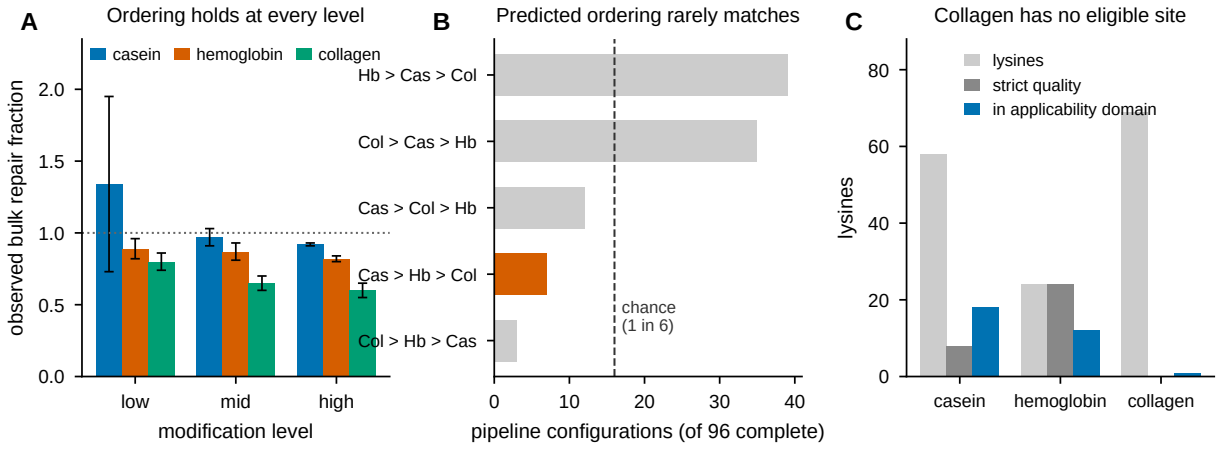


Figure S1: Protein-level comparison. (A) Observed bulk repair fraction by scaffold and modification level, with standard deviations over four replicates; the dotted line marks complete repair. (B) Distribution of predicted scaffold orderings across the 96 complete pipeline configurations; the observed ordering is highlighted and the dashed line marks the chance expectation. Cas, casein; Hb, haemoglobin; Col, collagen. (C) Lysines per scaffold, those meeting the strict structure-quality criterion, and those inside the applicability domain.

2 Response dispersion by substrate load

Dispersion is comparable across loads, with standard deviations spanning 0.298–0.357 and interquartile ranges 0.492–0.574, a difference of at most twenty percent between the widest

Table S1: Response distribution for the site sets analysed in the main text.

Load (ng)	n	Mean	SD	Min	Max	IQR	≤ 0.10 / ≥ 0.90
250	22	0.547	0.357	0.00	0.96	0.574	5 / 5
500	33	0.602	0.310	0.00	1.00	0.509	3 / 7
1000	37	0.479	0.304	0.00	0.89	0.533	7 / 0
2000	38	0.455	0.298	0.00	0.97	0.492	6 / 1

and narrowest condition. Mean response declines with increasing substrate load, the direction expected when enzyme concentration is held constant.

3 Descriptor extraction and validation

Structural descriptors were extracted for all 44 lysines appearing in the union across loads. The descriptor table distributed with the original analysis covers only the 33 sites of the 500 ng condition. To verify that our extraction reproduces that pipeline, descriptors were recomputed for those 33 sites and compared field by field. All thirteen fields agreed to floating-point precision, the largest absolute difference being 1.4×10^{-14} (residue solvent-accessible surface area). The eleven additional lysines (positions 36, 44, 151, 160, 228, 235, 248, 285, 340, 544 and 548) are ordinary lysines in chain A of PDB 4F5S and yield descriptors under the same rules.